3863. Minimum Operations to Sort a String

Solved

Medium Topics Hint

You are given a string `s` consisting of lowercase English letters.

In one operation, you can select any **substring** of `s` that is **not** the entire string and **sort** it in **non-descending alphabetical** order.

Return the **minimum** number of operations required to make `s` sorted in **non-descending** order. If it is not possible, return -1.

Example 1:

> **Input:** s = "dog"
>
> **Output:** 1
>
> **Explanation:**
>
> - Sort substring `"og"` to `"go"`.
> - Now, `s = "dgo"`, which is sorted in ascending order. Thus, the answer is 1.

Example 2:

> **Input:** s = "card"
>
> **Output:** 2
>
> **Explanation:**
>
> - Sort substring `"car"` to `"acr"`, so `s = "acrd"`.
> - Sort substring `"rd"` to `"dr"`, making `s = "acdr"`, which is sorted in ascending order. Thus, the answer is 2.

Example 3:

> **Input:** s = "gf"
>
> **Output:** -1
>
> **Explanation:**
>
> - It is impossible to sort `s` under the given constraints. Thus, the answer is -1.

Constraints:

- `1 <= s.length <= 10^5`
- `s` consists of only lowercase English letters.

Seen this question in a real interview before? 1/6

Yes No

Accepted 102,851/262.3K | Acceptance Rate 39.2%

Topics

Hint 1

Hint 2

Hint 3

Hint 4

Discussion (204)

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